In it for the Long Haul: Occupational Choice for Blue-collar Workers

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Abstract

In 2021, the American Trucking Association claimed that the driver shortage reached approximately 80,000 drivers and predicted a shortage of 160,000 drivers by 2030. This perceived driver shortage is a major concern for both policymakers and firms. In the 2021 Infrastructure bill, a trial program for a new policy was announced. This trial program authorized 18-to-20-year-olds to drive commercial vehicles. In this paper, I use a discrete choice mixed logit model to estimate the impact of different policies from various stakeholders on improving truck driver supply. These policies include wage increases, education cost reductions, and a policy of opening truck driving to 18-to-20-year-olds. I employ a control function approach to estimate the model, addressing endogeneity in wages and training costs resulting from oligopsony power. I use four data sets, the Current Population Survey March Supplement (ASEC), for individual and market share data. State-specific occupational wage data was sourced from the Occupational Employment Wage Survey (OEWS), work setting variables were sourced from O*Net, and educational costs were sourced from the Integrated Postsecondary Education Data System (IPEDS). The results show that higher wages, reduced training costs, and the inclusion of 18-to-20-year-olds increase the number of truck drivers. The largest potential increases come from adding 18-to-20-year-olds into the market, increasing trucking employment by about 46,000 new drivers nationally. JEL Codes: L91, J24, J44

Keywords: Truck Driving, Driver Shortage, Mixed Logit

1 Introduction

Truck driving employs approximately 3.5 million workers annually and is an essential industry in the U.S. economy. In comparison to other major transport modes, rail and water modes account for 65% of all tonnage and 70% of freight value when used as the sole mode of transportation. This makes US supply chains critically linked to the availability of drivers. Research in both the critical determinants of supply chain resilience and disruptions is growing, for example Elliott and Golub (2022), Wahdat and Lusk (2023), and Ferguson and Ubilava (2022). Industry stakeholders have raised persistent concerns about driver shortages. The American Trucking Association estimates that current shortfalls of 80,000 drivers annually will grow to 160,000 by 2030 without policy intervention.

These concerns have prompted substantial policy responses. The federal government allocated funds in 2023 to expand commercial driver's license (CDL) training programs and streamline licensing procedures. The 2021 Infrastructure Bill authorized a pilot program allowing 18-20 year-olds to drive commercial vehicles interstate under strict safety conditions, reversing a longstanding prohibition. Additionally, to facilitate easy entry into the occupation, there is a directive to standardize the CDL process across states. These initiatives reflect policymakers' assessment that expanding the eligible workforce is necessary to meet freight demand.

Despite extensive policy attention and federal spending, a fundamental empirical question remains unresolved: how responsive is the number of willing truck drivers to various policy interventions? In this paper, I estimate the labor supply elasticities to assess the increase in workers across three policy counterfactuals and the impacts of an aging workforce. Using a structural model of occupational choice and simulating counterfactual policies, I can make clear predictions on which policies can increase the number of workers.

I estimate a mixed logit model of occupational choice among blue-collar workers using individual-level data. The model allows for heterogeneous preferences over wages and occupation characteristics, with random coefficients varying by worker age and gender. To address endogeneity from local labor market conditions and employer market power, I employ a control function approach using Hausman instruments based on wages in neighboring states. I combine four datasets: the Current Population Survey for individual occupational choices, the Occupational Employment and Wage

Statistics for state-occupation wage data, O*Net for occupation characteristics, and the Integrated Postsecondary Education Data System for CDL training costs. The sample covers 2010-2019 and includes approximately 95,000 blue-collar workers across 48 states. The results demonstrate that truck driver supply responds to policy interventions but with considerable heterogeneity across policy types and demographic groups. The estimated own-wage elasticity is 0.006, implying that a $1,000 increase in annual wages attracts approximately 21,000 additional drivers nationally, a 0.6% increase. Sign-on bonuses targeting new entrants are less effective, generating roughly 15,000 new drivers but at a lower cost. Reducing CDL training costs by $1,000 adds approximately 7,000 drivers. In contrast, eliminating age restrictions for 18-20 year-olds could increase driver supply by 46,000 workers, a 1.3% increase in the supply of truck drivers. This suggests that regulatory barriers are a more fruitful avenue for policy change. State-level heterogeneity is substantial, with responses varying widely across states.

These findings have several implications for the labor shortage debate. First, truck drivers exhibit higher occupational retention (67%) than the average blue-collar worker (61%), contradicting industry narratives about retention crises. The binding constraint appears to be recruitment rather than retention. Second, the strong response to age eligibility changes demonstrates that regulatory barriers can have a significant impact, even in otherwise competitive markets.

This paper contributes to three literatures. First, it provides comprehensive estimates of truck driver labor supply elasticities using a structural occupational choice model. Second, I explicitly address the monopsonistic/oligopsonistic power in blue-collar labor markets, allowing me to credibly estimate wage elasticities. Third, it demonstrates the importance of modeling rich preference heterogeneity when evaluating policies targeting specific demographic groups. The remainder of this paper proceeds as follows. Section 1 discusses the economic definition of labor shortages and how the term often diverges from its economic meaning, as well as institutional details relating to the trucking industry. Section 2 reviews related literature. Section 3 describes the data and presents descriptive evidence. Section 4 presents the mixed logit model and identification strategy. Section 5 reports estimation results. Section 6 presents counterfactual policy simulations. Section 7 concludes.

1.1 Labor Shortages: Definitions and Implications

A shortage of drivers is a perennial concern for both firms in the trucking industry and the government. Government action has taken the form of various policies and grants. For example, in 2023 $48 million in grant funding was allocated to increase commercial driver's license (CDL) driver training opportunities, and $44 million of that figure was directed at simplifying and standardizing the process of attaining a CDL.[1] Despite the widespread acceptance among policymakers and firms that a shortage exists, it remains unclear whether there is indeed a shortage of truck drivers. In the context in which the term "shortage" is used, it lacks a formal definition. Often, when the term is used by Congress, lobbyists, and the press in discussions about the industry, it is not clear what each of these parties means, or if they mean the same thing. Furthermore, the shortage is forward-looking. A standard narrative is structured this way: "If there is no shortage today, then there will be one in the future due to any one of several factors." Conversely, many people believe that human truck drivers will be entirely supplanted by AI, so even if there is a shortage, it won't matter in the long run. Since all of these concepts are bundled into one word, the concept becomes extremely fluid in its usage and interpretation. To clarify the discussion, I will mention some of the various explanations for the shortage, drawing on other similar labor shortages that economists have historically focused on. And finally, in this section, I will define 'shortage' and explain how I utilize it in this paper, and describe why I believe this definition is appropriate.

The classic model of supply and demand yields the most common definition of a shortage. In this context, a shortage implies that the number of drivers supplied in the market at the prevailing wage rate is insufficient to meet the quantity of drivers demanded. In a labor supply and demand model, this would occur if a barrier prevents the wage rate from rising to the equilibrium level. An example of such a barrier is a maximum wage law.[2] Cappelli (2015) cites a skill gap as the barrier to equilibrium employment. In this case, the concern would be that we don't have enough drivers because there aren't enough skilled workers to drive. He finds little evidence of a skills gap in the US economy, but rather, firms prefer to lower labor costs. Alternatively, the labor supply curve

[1] Announcement: https://www.fmcsa.dot.gov/newsroom/biden-harris-administration-announces-grants-improve-commercial-drivers-licensing-process

[2] There are no immediately obvious structural barriers that might act like a maximum wage. One possible explanation is the CDL requirements, which exclude a portion of the market. However, these requirements are generally fairly low-cost to acquire, and the estimates indicate that the impact of lowering these costs is also small.

could be inelastic, meaning small wage increases induce only a few workers to move occupations. As I show later, workers respond to wage increases by shifting into occupations that are now higher-paying, although my estimated elasticities are low. The absence of evidence is not evidence of absence, but in this circumstance, it seems unlikely that an economic definition of shortage applies well to this market.

Notably, the discourse regarding the truck driver shortage is prima facie similar to that of the nursing shortage. The introduction of Shields (2004) is extremely similar to the motivation in this paper, except it is in the context of nursing. Shields (2004) claims there was a shortage of 126,000 whole-time-equivalent nursing positions in 2001, and the shortage was expected to increase to around one million by 2010. While the numbers in nursing differ, the underlying motivation remains the same. There is a shortage today; it is bad, but it will worsen tomorrow if we don't address it. Shields (2004) also cites aging demographics as a potential cause of a shortage. Like the trucking industry, policymakers are concerned that an aging workforce will exacerbate a nursing shortage.

The nursing shortage contains a few more valuable points to contextualize the trucking shortage. Furthermore, a significant amount of high-quality work has been done analyzing the nursing shortage. Sullivan (1989) proposes that the nursing profession is a great example of monopsony power in the labor market context. Sullivan (1989) summarizes the debate around monopsony power and a nursing shortage. Hospitals are large employers of nurses; about 30%[3] of all registered nurses are employed by hospitals. Furthermore, nursing is characterized by geographic segregation and a significant regulatory burden. Within a geographic region, there may be only a few employers for nurses and a hospital or two. In general, hospitals serve a local area, and this feature would lend itself to imperfect competition, especially if a subset of the labor market is unwilling to relocate.

Furthermore, the regulatory burden reduces the dynamism of the market, and the high costs of entry give incumbents an advantage. Trucking companies and workers face regulatory burdens similar to those of nursing, albeit the financial expenses for becoming a licensed driver are much lower than those of nursing. However, drivers and nurses are bound by professional standards and are regulated for safety. The consequences for violating the standards are similar and include permanent exclusion from the profession for violations. For example, truckers face both imprisonment

[3] Quarterly Census of Employment and Wages, Bureau of Labor Statistics (2023)

and permanent revocation of their CDL for engaging in human trafficking, just one such example. Because of these factors, monopsony power appears as an attractive explanation for trucking, like nursing. However, monopsony power is not the only explanation for a shortage. In fact, despite the notional similarities, a few features make the explanation less likely. First, trucking companies operate in much larger geographic areas, meaning any individual driver has many options. As aforementioned, there is a very low cost for an individual driver to become a sole proprietor.

Long et al. (2008) lays out a variety of alternative explanations for the nursing shortage. The first three are standard explanations in the literature, and the final three are novel. While their paper serves as a detailed explanation of all six, I will mention a few here. First, they observe that the nursing shortage is often a retention crisis. It's not that there are too few nurses; rather, they don't stay employed. The first explanation proposed is that the shortage is not a shortage, at least not in the economic sense. Hospitals have high turnover rates for nurses, potentially evidence of a shortage. However, short-term nurses are a close substitute for permanent nurses. When hospitals hire these short-term nurses, this behavior represents profit-maximizing behavior on behalf of the hospitals. Since both nurses and hospitals are engaged in profit-maximizing behavior, there is no economic shortage despite hospitals having high turnover. The labor market is in equilibrium as in any other supply and demand model. Whether this behavior is good for consumers is another question. The turnover-shortage explanation can be applied in trucking with a different underlying mechanism. Min and Lambert (2002) cites the high turnover of truck drivers as a decision made by the drivers, not the firms. Like nursing, they suppose the shortage is caused by the low retention of truck drivers in firms. However, if we apply the logic from Long et al. (2008), the drivers are profit-maximizing, and note that many 'driving' occupations are close substitutes for truck drivers. It seems unlikely that turnover is the evidence or the source of an economic shortage in trucking. Furthermore, if we include trucking firms in the analysis, we get a similar conclusion. While firms indicate they would prefer workers to stay for longer tenures than most workers, it is not clear that the firms are not behaving rationally and hiring the appropriate workers at the prevailing wage. In fact, the retention crisis is cited as a significant component of the driver shortage (ATRI, 2023)[4]. Firms could, in principle, increase the benefits of working by increasing tenure. The reality is that

[4]American Transportation Research Institute publishes an annual report on major concerns in the trucking industry. Their sample consists of stakeholders representing both firms and drivers. Link: https://truckingresearch.org/about-atri/atri-research/top-industry-issues/

there is a marked difference between what firms say and what they do. If firms are not paying more for retention, it is unlikely that they would pay more because higher pay would lower their profits. Following this logic chain to its conclusion, as we did with nursing, implies that the labor market is in equilibrium. We are also concerned about consumers downstream. One may believe consumers are worse off under this scenario because of the mismatch between drivers and loads. However, like wages, the price of shipping goods could increase, and more drivers would, in principle, enter the market. Secondly, turnover does not necessarily imply a shortage in the same way it would in nursing. Self-employment is also an attractive option since the trucking industry consists of several firms.

The second explanation for the nursing shortage focuses on funds-constrained hospitals. The core idea is that there are shortages at hospitals that cannot afford to pay premium wages. A hospital could be funds-constrained because of its tax status (non-profit), local market conditions, or management. This explanation is attractive in trucking because some firms may have difficulty retaining workers, and it might be the case that those firms are poorly managed or are engaged in an extremely competitive market. However, this is not necessarily a type shortage that policymakers are worried about since this is localized to specific firms.

The final proposed explanation is based on professional standards. In essence, the shortage is caused by regulatory barriers. This mirrors my concern about the trucking industry regarding the difference between an economic shortage and a supply chain shortage. It might be better for firms if the barriers to becoming a driver were lower; however, the reason these barriers exist in the case of trucking is that there are externalities associated with licensing people below a certain threshold. It is unlikely that firms would hire these workers if they had to pay the full cost of employment.

There is one final dimension to discuss regarding nursing versus trucking. Nursing is a highly female-dominated field, and trucking is a male-dominated field. Both are occupations with large gender gaps. This difference provides one tool to address a shortage. Either of the two sexes heavily dominates both occupations. Therefore, each occupation would likely benefit from recruiting more individuals of the non-dominant sex. Alternatively, there may be structural or societal reasons why the occupations are sex-segregated, and addressing these limitations would broaden the work pool. In the case of truck driving, this is not dissimilar from removing legal barriers and opening the market to 18-20-year-old drivers, a segment of the market previously barred from entry for legal reasons. Addressing barriers to men and women in these fields would open more opportunities to hire drivers.

Although the list of definitions or causes for a shortage I reference is not exhaustive. The definition I use in the paper is a non-economist definition of shortage defined as: *The number of drivers supplied in the market does not meet the number of drivers necessary to cover the total logistic load at the equilibrium price.* Logistic load, as I define it here, is the number of loads shippers want serviced at any given time. Essentially, some shippers are willing to wait or unwilling to pay the prevailing cost for shipping freight; consequently, the load is delayed. These are the shippers whose willingness to pay is less than the cost of freight services. To make it more concrete, if three firms would like freight carried, there are three carriers to meet that demand; if the final match couldn't agree on a price, we would have a shortage. Since the load is not carried at that price, this definition is more plausible than an economic shortage, since there are no apparent market failures under the status quo in the labor market for truck drivers. Secondly, this allows room for improvement; increasing the number of drivers can, in principle, reduce the strain on the logistics system and fortify the transportation market against shocks. While I address moving along the labor supply curve in the form of elasticities, the most impactful policy is an outward shift in labor supply.

In the context of the classic model of supply and demand. Consumers are bifurcated by the equilibrium price of the good into the buyers whose willingness to pay is at or below the market price and consumers unwilling to buy at the price. In the same way, some firms would be willing to ship their product if the price were lower, but drivers are unwilling to take loads at that rate. Thus, some portion of the total logistic load goes unserviced, but the market is in an equilibrium without a shortage in the economic sense. This fact is not at odds with a market being in equilibrium. Fundamentally, this explanation of the shortage states that the shortage results from firms being unable to hire workers with a higher reservation wage.

1.2 Institutional Details: Policies for the trucking industry

Presuming that the current number of drivers is insufficient, there are two margins we can modify to increase the labor supply of drivers. The first is the intensive margin; in principle, we can increase the amount of loads serviced by increasing the number of hours a driver is willing or able to work. While I do not consider any of these policies in this model,[5] discussing some of the potential policies here is worthwhile. Currently, drivers are regulated by the number of hours they are allowed to be on duty. This policy is in place for obvious safety reasons, but it also is a constraint that binds drivers labor hours. A policy relaxing the constraint would increase the amount of labor supplied. The apparent trade-off between safety and work hours is a clear example of the role of public policy. Alternatively, there is an ongoing discussion of truck parking, its availability, and the role of parking constraints in reducing available hours for drivers. Parking is a scarce resource and is distributed in discrete intervals that don't necessarily match the time constraints of drivers. Furthermore, the gap between parking demand and drivers on busier corridors is stark. Consider a driver faced with the following dilemma. If the driver has 4 hours remaining and sees an open truck stop, they must choose whether to forgo the 4 hours they have remaining and occupy the nearest truck stop. Or they use their remaining 4 hours and are not guaranteed an open slot at the following parking location, potentially parking illegally. The driver must choose whether to stop or risk it. In particular, the drivers will be concerned about the likelihood that the next truck stop is at capacity, because if it is, they will be stuck on an exit without facilities. Private solutions are now available. Services such as Truck Parking Club[6] is a popular example providing matches between drivers looking for parking and landowners with open lots. These services link landowners with drivers to provide parking in safe locations, mitigating the capacity constraint at truck stops. The Bipartisan Infrastructure Bill allocated funds to increase truck parking in certain states[7]. Improving technology and reducing wait times for drivers is another way to increase the amount of freight they can carry. The driver's time on the road can be used up by dwell time, stuck waiting for cargo to be loaded or unloaded. Reducing dwell times is one way to increase the amount of freight drivers are able to carry. Since dwell times are a form of externality, the warehouse does not necessarily pay the full cost imposed on a driver for delay; a tariff system could be utilized to shift the costs from the drivers to the warehouse, aligning incentives.

On the extensive margin, there are more policy proposals. The popular press has fixated on the concept of a shortage; a Google search yields a steady stream of articles over time from generally

[5] The model works by modeling the extensive margin for truck driver supply. Other models could look at the intensive margin, which would increase the number of hours a worker could work.

[6] Truck Parking Club Website: https://truckparkingclub.com/

[7] Announcement: link

reliable sources, such as The New York Times, NPR, CNBC, and Time Magazine, discussing the causes and potential cures of the truck driver shortage. This widespread concern about a shortage led the Biden Administration to push for several programs, including the 2021 Bipartisan Infrastructure Bill, which authorized a pilot program to trial a policy that would increase the supply of truck drivers.

The pilot program has many features I don't model directly in this analysis. Because it is a pilot program, firms must meet specific criteria to qualify as eligible. Second, firms do not automatically opt in and must register with the DOT. Firms qualifying for the program must keep their vehicles governed at or below 65 miles an hour, meet stringent safety ratings, and not carry high-risk cargo such as hazardous materials or double/triple trailers. The policy modeled in this paper allows 18-to-20-year-olds to work as drivers in the same way that drivers who are 21 and older can. Essentially, the policy opens the market and makes the restrictions similar to other blue-collar occupations.

Other important policies targeting new drivers include expediting the CDL process across states, advertising programs intended to recruit veterans, and addressing the working conditions of truck drivers. These policies are a mix of government policies, such as standardizing the CDL process across states, and firm-specific policies, like addressing working conditions.

The extensive margin refers to the increase in the number of drivers in the market, and this is where I focus my analysis. I include four counterfactuals that address different policy proposals and a fifth that considers the impact of an aging workforce on the share of drivers. The first three pecuniary counterfactuals are a wage increase for all potential and incumbent truck drivers, a sign-on bonus for only new entrants, and a reduction in the training cost of CDL programs. The fourth is allowing 18-20-year-olds to choose truck driving as a profession. Finally, the final counterfactual considers the impact of an aging workforce and what happens to the market share of truck drivers in the blue-collar labor force as the workforce ages.

2 Literature

Discrete choice models have a long history and many applications. Introduced by McFadden (1974), the conditional logit model proved to be a tractable and empirically useful way to model many choices. He demonstrated the power of the logit choice model in the context of consumers' shopping decisions. In this paper, I use a product demand model applied to occupational choice. Since I have individual-level data available, I model the individual decisions in the style of Goolsbee and Petrin (2004). They model a household's choice of television provider using a mixed logit model to compute the welfare gains from the addition of satellite television providers. The convenience of the approach allows the researcher to use individual-level data and aggregate the results to market outcomes. Alternatively, it is possible to estimate the model using aggregate market outcomes with generalized method of moments, Berry et al. (1995), Nevo (2001), and Petrin (2002).[8]

Boskin (1974) was one of the earliest papers to model occupational choice as a discrete choice problem. Comparing 11 occupations in the context of a conditional logit model, individuals choose their occupation based on the present value of future earnings in each occupation. Other models have focused on the dynamics of work, education, and leisure choices of workers. Keane and Wolpin (1997) use a dynamic discrete choice model to capture young men's human capital investment and occupational choice. Fuller et al. (1982) model students' college choice as a function of both the student's characteristics and the economic costs of the school. I use a basic version of this to model the cost of a CDL training program.

Phares and Balthrop (2022) uses a mixed logit model to estimate the labor supply of truck drivers in industry occupation pairs. Their methodology allows for rich substitution patterns between occupations, but they restrict the choice set to five possible alternatives, including truck driving. Importantly, they compute the wage elasticity for truck drivers. They show that a 1% increase in the wage rate corresponds to a .3% increase in drivers. This result suggests that drivers do respond to wage increases. Other work by Richards and Rutledge (2022) uses a dynamic search model to estimate the impact of the COVID-19 pandemic on truck rates and labor supply. Their results suggest that the trucker market is in disequilibrium since the creation of new jobs exceeds the rate at which the jobs are filled. They expressed concern that the delay between training and work may partially cause the disequilibrium.

[8] Although I don't estimate my model using this style, the earliest versions of the project were done using this methodology initially. I chose to use individual-level data due to the issues with observed zero market shares. This allowed the model to be computable.

3 Data

The data spans the period from 2010 to 2019. It is a combination of four public data sets: the Current Population Survey ASEC March Supplement (CPS), Occupational Employment Wage Statistics (OEWS), Integrated Post Secondary Data System (IPEDS), and O*Net Occupational Characteristics. The CPS records economic and job data for a sample of American households. The CPS data was downloaded through the Integrated Public Use Microdata Series (IPUMS) from the University of Minnesota Flood et al. (2023). The CPS allows me to observe a worker's previous and current occupations because I observe the worker twice a year apart. O*Net records hundreds of job characteristics for most occupations. OEWS is the source of all employment and wage data used in the analysis. Finally, IPEDS is the source of truck driver training programs by state and year.

The data includes a total of 69 occupations; 47 of the 69 occupations are blue-collar or white-collar with minimal education requirements.[9] I excluded 22 occupations that are either educated white-collar or agriculture occupations and have extremely small sample sizes. This restriction removes about one-third of the CPS sample. An exhaustive list of excluded occupations can be found in the appendix. However, some examples are advanced computer and Math Workers, Advanced Financial Workers, Advanced Medical Workers, Engineers and Architects, and Education Workers. These occupations tend to require advanced degrees such as an M.D. or master's degree or many years of labor market experience. I exclude these occupations for two reasons. First, my model is insufficient to explain the educational decision-making process of a worker pursuing an advanced degree. Dynamic discrete choice models exist that can capture the human capital accumulation versus work decision of workers Keane and Wolpin (1997). However, since I only observed each individual twice, I cannot track them over time and estimate the underlying utility function of a worker. In particular, I don't observe most workers during the major human capital-forming years. For example, a bachelor's degree can take 3-5 years, and more advanced degrees require more time. Because of this, most occupations in my sample take less than one year to train for. Given my data set, the 22 educated occupations are too complicated to accommodate. Secondly,

[9] The CPS data uses a coding system based on the 2010 Census Classification Scheme. O*Net uses a slightly different scheme with narrower categories. I created a crosswalk that simplifies the occupation into broader categories, resulting in 69 occupations.

the overlaps between my main occupation of interest and these higher education occupations are relatively small. This is a lucky accident because it minimizes the simplification of reality and preserves parsimony. About 80% of all educated white-collar workers in my sample are either not working or members of the educated white-collar category. The other 20% of educated white-collar occupations are distributed among the other 50 occupations, including agriculture. The other 68 occupations contribute a small share of workers to the total share of educated white-collar workers. In general, there is relatively little movement between these two groups.

I divide occupations into two classification schemes: major and minor occupations. The minor occupation codes are broadly based on the Standard Occupational Classification System (SOC). The multiple datasets I use are not internally consistent in how they classify occupations. Because each dataset I use uses a slightly different coding system, I create my own classification schemes. Luckily, both OEWS and O*Net use the Standard Occupational Classification System (SOC) code. However, O*NET uses the more modern 2019 codes, and OEWS uses the 2018 (SOC) codes. These two codes have significant overlap. However, the CPS uses a separate system based on the 2010 (SOC) Codes. I created my own occupational coding scheme to harmonize all these occupational systems. And reduce the dimensionality from the 997 unique occupations in the O*Net system to just 69. This process resulted in the 47 included and the 22 excluded occupations mentioned earlier. The major occupation codes comprise nine broad occupational classes based on the 69 occupations. I use seven codes: white-collar worker, tradesman, manual laborer, service worker, production worker, transportation worker, and sales worker.

The CPS ASEC March supplement is a survey that asks detailed questions about the workforce. Each person is sampled twice, one year apart, and if a person switches occupations between samples, the survey will reflect that change in occupation. Within a given sample year, half of the sample is incoming, and the other half is outgoing; I treat the observation year as the outgoing year. The incoming sample is used only to record their occupation, and that is my measure of the previous occupation, which is not a choice variable for the worker. A worker is endowed with their previous occupation and maximizes utility conditional on that occupation. The outgoing occupation is the dependent variable, the worker's choice of occupation. This takes the repeated panel structure and converts it to a repeated cross-section, where I observe a new set of individuals each year. But I have information on their current and previous occupations in that year. The CPS survey in the cross-sectional format with the updated occupation codes is the primary data source for the analysis. Aside from measuring occupational switching, the CPS data contains multiple demographic characteristics of each worker, including age and sex. I use age and sex as my main demographics in the analysis since two of the major concerns in the trucking industry are the aging workforce and the heavy male distribution. Recruiting more women is a potential source of new drivers, and understanding how women respond to truck driving's observable characteristics can shed light on the effectiveness of recruiting women into truck driving.

Most individuals in the sample do not change occupations (72.5% average retention). Table 1 shows the top and bottom ten occupations by retention in the sample. There is considerable heterogeneity across occupations in the retention rates, ranging from an average high of 80 % and a low of 18 %. Interestingly, truck drivers have one of the highest retention rates of all occupations in the sample. This suggests a policy that increases the number of truck drivers likely has to operate through recruiting new workers rather than retaining the current workers. Hazmat workers have the lowest retention rate in the sample; given the danger of the occupation, this is not necessarily surprising.

Table 1: Retention rate by Occupation

Retention Rate (H-L)	Occupation	Retention Rate (L-H)	Occupation
0.80	Emergency Service/Criminal Justice	0.18	Hazmat Worker
0.74	Personal Appearance Workers	0.28	Information Clerks
0.68	Housekeeping/Janitorial Worker	0.35	Repair Worker
0.67	Bus and Taxi Drivers	0.35	Forestry Worker
0.67	Truck Drivers	0.36	Unemployed
0.65	Electrician	0.38	Gaming Service Worker
0.65	Courier	0.39	Laundry Workers
0.64	Rail Workers	0.40	Industrial Repair Worker
0.64	Food Preparation Worker	0.41	Advanced Electronics/Precision Installer
0.62	Funeral Workers	0.41	Material Movers

Notes: Average retention rate by occupation in the sample. Calculated using the CPS by computing the number of people in the sample who are no longer incumbent in their previous occupation when observed in the outgoing rotation.

O*Net data consists of 877 occupations and corresponding characteristics in six general categories called the Content Model. The six categories are worker characteristics, worker requirements, experience requirements, occupation requirements, occupation-specific information, and occupation characteristics. I use a subset of occupational requirements, denoted as work context. The work context category is defined as the physical and social factors that influence the nature of work. These features describe the working conditions and pressures of an occupation. The characteristics are unchanging over time, firms cannot change the occupation characteristics. The easiest way to conceptualize these characteristics is to consider them as if they are product characteristics in product choice models. In many ways, these occupation characteristics are the corollary of product characteristics in the context of this paper. For example, features of products, such as mpg, color, flavor, etc, are very similar to the eight variables I use to distinguish work conditions. These characteristics are the consequence of error, contact with others, degree of automation, exposed to hazardous conditions in an enclosed vehicle or equipment, outdoors exposed to weather, time spent using your hands, and time pressure. These variables are time constants and measure the characteristics of a particular occupation. Since I observe the current and previous occupations, I calculate the Euclidean distance in characteristics between the two occupations. The more similar an occupation, the closer they are in distance, and the more dissimilar occupations are, the more considerable the distance.

OEWS is the source of market wages and employment. I measure the expected compensation using the median and the 25th percentile full-time wage income. All market wages are converted to 2019 dollars using the Consumer Price Index (CPI). I do not distinguish between part-time and full-time workers since I am modeling the choice of occupation, not the number of hours a worker chooses to work. I use different measures of occupation wages to capture the differences in wages for incumbent and new workers. I use the median full-time annual income for incumbent workers to measure market wages when they consider their incumbent occupation. However, if a worker considers an occupation they do not currently occupy, they would observe the 25th percentile of annual income as their expected annual income.

IPEDS is the source of education training data for CDL programs in the United States. While all publicly funded institutions are required to submit their data to IPEDS, many training programs are missing from the data. For example, in 2019, 10 states were missing from the dataset.[10] In any given year for the states with at least one truck driving training program, I compute the mean of the training cost by averaging the observed costs of the training program for each school observed within the state and year. I use the average training cost within each state as the observed training cost in that state.

[10]The missing states are Colorado, Connecticut, Louisiana, Maine, Massachusetts, Nevada, New Jersey, Rhode Island, Vermont, and Wyoming.

The final dataset comprises 48 states over nine years, excluding Alaska and Hawaii. There are three categories of variables. The individual-level data from the CPS varies across individuals and includes age, sex, and previous occupations. The State-Time variables vary across states and time, including expected wage and training costs. Finally, the occupation characteristics only vary across occupations.

The summary statistics are below in Table 2.

Table 2: Summary Statistics

	Female		Age		Retention Rate		Distance		Earnings		Training
	Sample	Trucker	Sample	Trucker	Sample	Trucker	Sample	Trucker	25th Pctl	Median	CDL
Mean	0.40	0.04	46.5	49.1	0.61	0.68	2.09	2.87	$36,912.90	$46,611.33	$4,537.27
Std Dev	0.49	0.20	13.4	12.4	0.49	0.47	0.97	1.10	$15,022.12	$20,135.72	$2,959.29
25th Pctl	0	0	36	40	0	0	1.27	2.04	$25,496.75	$31,294.78	$2,461.42
50th Pctl	0	0	47	50	1	1	1.98	2.54	$33,569.24	$42,430.56	$3,588.63
75th Pctl	1	0	57	58	1	1	2.73	3.84	$45,815.06	$58,535.18	$5,364.65
Obs	95,496	3,966	95,496	3,966	95,496	3,966	37,663	1,264	35,744	35,744	380

Notes: Summary statistics computed from the four data sets. Female, Age, Retention Rate and Distance are all calculated from the CPS. Earnings are computed from the OEWS, and Training costs come from IPEDS.

Table 2 contains the various demographics of the entire sample and truck drivers and the earnings distribution over occupations. There are 103,399 total individuals in the sample, of which 4,061 are truck drivers. About 3.9% of the sample consists of truck drivers, and the other 96.1% of workers are employed in different occupations. No workers in the sample are unemployed. The age, percent female, and retention rate variables are calculated for the entire sample and truck drivers. The distance columns were calculated only for movers. Since all non-movers move zero distance, and about 78% of the sample stay in their previous occupation, computing the summary statistics with the non-movers is less informative. The statistics about how far movers move are more informative because they give some measure of where workers go when they leave their incumbent occupation. The annual earnings statistics are calculated from the state and year variation in real annual earnings across all occupations. Each incumbent worker will see the median annual earnings as their predicted wage for staying in their current occupation, and they will observe the 25th percentile for all other occupations.

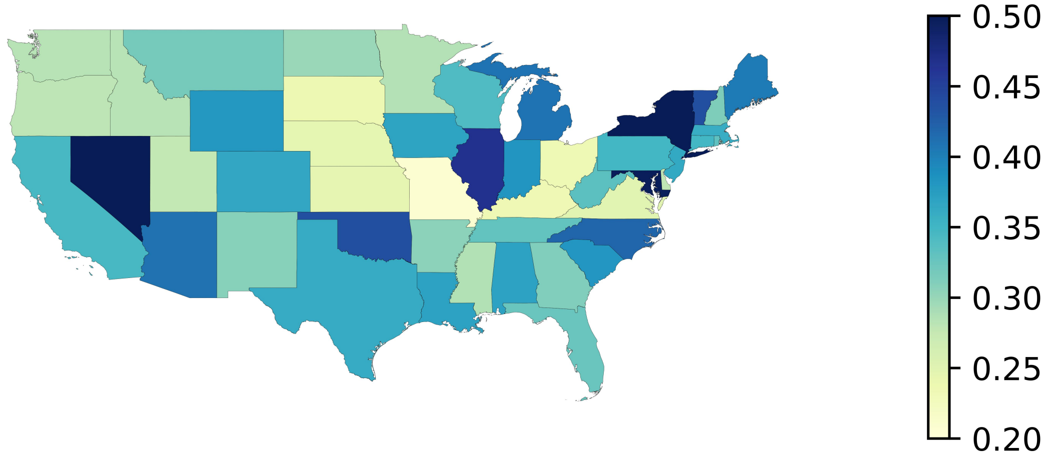
The driver sample is both 2.6 years older on average and significantly more male than the average worker. This is a well-documented feature of the truck-driving industry; many are concerned about an aging male-dominated workforce. The average age of truck drivers is 49.1, slightly higher than the sample average of 46.5. However, this difference is much less pronounced than the ratio between men and women in the sample. The share of women in truck driving is very small; only about 4% of the trucker sample is female compared to about 40% in the whole sample. The sample, on average, skewed more toward males, and truck driving is even more skewed. Truck driving has a higher retention rate than the sample average. Considering this information in the context of a presumed worker shortage, the relatively high retention rate within the occupation is surprising but does not necessarily contraindicate a shortage.

Annual earnings are the real annual earnings converted into 2021 dollars, although the sample ends in March 2020. Each occupation is observed in the OEWS by state and year. To correct the role of experience and tenure in an occupation, I use the median wage for tenured workers and the 25th percentile for new workers. This rule implies that new entrants are new or not experienced enough to command the higher wage. So, for example, using the numbers from the mean occupation, an incumbent worker would observe annual predicted earnings of about $47,000 if they were to stay in this fictional mean occupation. A potential entrant would observe $37,000 because they are new to the occupation. The occupations in the sample are not exceptionally high-paying occupations; About 75% of occupations have a median earning below $59,000.

The CDL training program costs are observed at the state level across multiple years. However, due to data and computational limitations, I only used 2019 to estimate the model at the national level, using cross-sectional variation. In terms of variation, the median program costs about $3,600 to complete, with a large standard deviation across states. This is for two pragmatic reasons. The first reason is my use of Google Colab to speed up the estimation process. The model takes time to converge, and using Google Colab helps speed up the process, but they have data limits on allowed RAM, which the complete data set exceeds. The second is that the maximum likelihood estimation I use struggles in practice with the unbalanced cross-section. In principle, the estimation should be fine, but convergence can fail even with subsets of the whole period.

Figure 1 shows the geographic distribution of new truck drivers. The figure is a choropleth showing the share of new drivers within that state. The dark states have the largest share of the current truck drivers as recent entrants, and the lighter states have fewer entrants. For example, a value of 1 in West Virginia would indicate that all drivers in the sample from West Virginia are new entrants. New York, Nevada, Illinois, and Maryland have the largest share of new entrants

Figure 1: Share of New Truck Drivers by State



Notes: The share of new drivers by a value of 1 would indicate all drivers in the sample in a given state are new. This figure uses all sample years to compute the flow.

to incumbent drivers. This means these states have the highest rate of entry compared to other states.

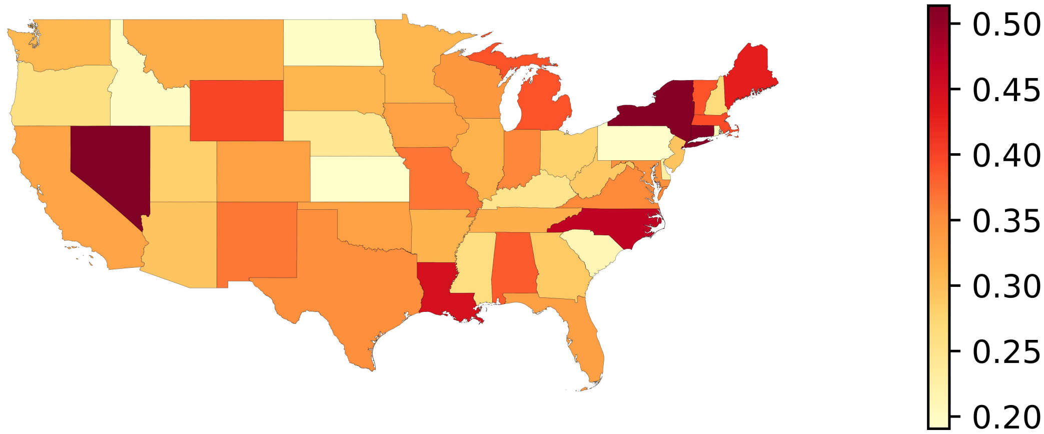
Figure 2 shows the geographic distribution of former truck drivers. It is the exit rate by state in the truck-driving occupation. Nevada, Maine, North Carolina, Louisiana, Vermont, Boston, Michigan, and Wyoming all have the lowest retention rates. For example, a value of .3 means 30% of past truck drivers are no longer truck drivers. Pairing Figure 1 and Figure 2, we can see that Nevada, New York, and to some degree North Carolina have a significant degree of churning in the labor market for truck drivers. Within these states, we have a high share of new entrants and a high share of people leaving the occupations. In contrast, Illinois and Maryland appear to be net gainers, having a large share of new drivers but a low share of exciters relative to their increases. There appears to be no clear geographic pattern other than a high degree of churning in some states like Nevada and New York.

4 Model

4.1 Workers

Using a mixed logit framework, I model the number of workers in an occupation as an aggregate outcome of many heterogeneous individual decisions about their job options. The mixed

Figure 2: Share of Former Truck Drivers by State



Notes: The share of former drivers by a value of 1 would indicate all drivers in the sample in the previous year in a given state quit being drivers. This figure uses all sample years to compute the average change.

logit approach offers more flexibility than conditional or multinomial logit alternatives, or modeling market shares of each occupation directly. Since individual-level data is available, I use individual decisions instead of aggregating them up to market outcomes. In this case, the mixed logit is particularly attractive because of the heterogeneity in preferences across workers. Workers in different segments of the population will have different preferences. Furthermore, the demographics enable me to make inferences about the aging workforce and the potential for recruiting younger workers. The complex substitution patterns across occupations also lend credibility to the precuniary counterfactuals, which are richer when accounting for heterogeneity across workers.

A worker's choice probability of a given occupation depends on the characteristics of the past and current occupations, wages, education cost, worker demographics, and an idiosyncratic Gumbel taste shock. Workers take their past occupation as given and choose among 68 possible occupations, excluding their own. A worker observes the wage rate they would earn w_{js} if they select occupation j. If they are incumbent in occupation j, they observe the median wage rate of occupation j; otherwise, they observe the $25th$ percentile. The distance between occupations is calculated as the Euclidean distance between occupations in the characteristics space. Large values of $||x_j, x_k||$ indicate very different occupations, and smaller values indicate more similar occupations. Naturally, when $j = k$, the distance measure is zero. The training costs are only observed for people considering truck driving as an occupation but are not already incumbent drivers. The training cost is measured as the average cost of a CDL training program in the driver's observed state. I consider two separate utility specifications, one national, which includes truck driving training program costs as a component of an individual's choice probability. The other utility specification is estimated state by state, allowing for more heterogeneity and variation in preferences by state. There is a tradeoff between these two specifications. The state-by-state specification does not allow for the inclusion of the cost of truck-driving training programs because the variable varies across states. The national model compromises by using only a single year of data and not fully leveraging variation over time.[11] However, it allows for more flexibility. The national-level utility function is specified as follows:

$$u_{ijk} = (w_{js} - \mathrm{FC}_s)\alpha_i + ||x_j, x_k||\beta_i + \xi_m + \varepsilon_{ijk} \tag{1}$$

Since the training cost and annual salary are pecuniary, I deduct the cost of a truck-driving training program from the expected annual salary of a new entrant. The new variable is the annual total compensation for choosing truck driving as a profession. All occupations other than truck driving don't have an observed training cost, so the total compensation is just the annual wage rate. The training cost is an unobservable for all occupations other than truck driving. I will discuss the control function approach in a later section.

The second model, which I use for three of the four counterfactuals, allows richer heterogeneity across states, indicated by subscript s. The utility of job j for a worker i given their previous occupation j and state s:

$$u_{ijks} = w_{js}\alpha_{is} + ||x_j, x_k||\beta_{is} + \xi_{ms} + \varepsilon_{ijks} \tag{2}$$

Where

$$\begin{pmatrix} \alpha_{is} \\ \beta_{is} \end{pmatrix} = \begin{pmatrix} \alpha_s \\ \beta_s \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \beta_{Fs} & \beta_{As} \end{pmatrix} \begin{pmatrix} \mathbf{1}(Female = True) \\ Age \end{pmatrix} + N(0, \Sigma_s)$$

Both the national and state-by-state models use this random coefficient structure. The only difference between the state-by-state and national models is the state-specific standard deviations on

[11] This is a compromise because of computability. The most general model would utilize all data sources and employ the specification of the state model; however, the full model is too large for both Google Colab and my desktop.

the random coefficients and mean values of coefficients.

The wage response of worker i depends on the mean wage response in state s and individual specific draw a normal distribution with mean zero and a standard deviation σ_s specific to state s. I capture all worker-level heterogeneity of wage responses within a state with the estimated standard deviation of the normal distribution. Workers' preferences on distance are more complicated; the preferences depend on a normal draw and the age and sex of a worker. β_{is} can be considered the disutility of adapting to a new occupation. For example, if a worker chooses to stay in occupation k, they would experience no disutility from moving. However, if a worker decides to switch occupations, they would experience disutility in accordance with their value of β_{is}. Presumably, workers vary in their tastes for differences in occupations; it is plausible that some people even experience positive utility from differences in occupations. Allowing the model to capture the heterogeneity of people's taste for differences is essential. In particular, if women and aging populations are reluctant to move occupations, this will have important consequences as the workforce ages and how effective recruiting women into truck driving will be.

The unobservable occupation characteristics are estimated at the major occupation code level. The wages and differences in occupations do not control for these characteristics. The ξ_{ms} for major code m is the mean utility for choosing an occupation within the major code m. ε_{ijks} is the unobservable factors influencing an individual's occupational choice. I assume ε_{ijks} follows the Gumbel distribution and is independently distributed across choices. I will discuss the unobservables contained in ε_{ijks} later.

The probability a worker i chooses an occupation j is:

$$L_{ij} = \frac{\exp\left(w_{js}\alpha_{is} + ||x_j, x_k||\beta_{is} + \xi_m\right)}{\sum_k^K \exp\left(w_{js}\alpha_{is} + ||x_j, x_k||\beta_{is} + \xi_m\right)} \tag{3}$$

Depending on the draws and characteristics of the worker, the probability a worker chooses a particular occupation changes. Across all individuals in the labor market, the observed share of workers in an occupation is σ_j:

$$\sigma_j = \int L_{ij} dF(\alpha_{is}, \beta_{is}|Age, Sex, \Sigma) \tag{4}$$

The aggregate share of workers in occupation j is the weighted sum of the choice probabilities using the probability density function of the parameters as weights. Using these models, I consider four counterfactuals to increase the number of truck drivers: a wage increase for all truck drivers, a signing bonus for new truck drivers, allowing workers aged 18-20 to drive trucks, and a reduction in training costs. I compute a fifth counterfactual to investigate the impact of an aging workforce and whether there will be a decline in the number of truck drivers as the workforce ages.

The wage and training cost elasticities are specified as follows:

$$e_{\sigma_j,w_\ell} = -\frac{w_\ell}{\sigma_j} \cdot \int \alpha_i L_{ij} L_{i\ell} \, dF(\alpha_{is}, \beta_{is} | Age, Sex, \Sigma) \tag{5}$$

$$e_{\sigma_j,w_j} = \frac{w_j}{\sigma_j} \cdot \int \alpha_i L_{ij} \cdot (1 - L_{ij}) d \, F(\alpha_{is}, \beta_{is} | Age, Sex, \Sigma) \tag{6}$$

$$e_{\sigma_j,\mathrm{FC}_\ell} = \frac{\mathrm{FC}_\ell}{\sigma_j} \cdot \int \alpha_i L_j(\beta) L_\ell(\beta) \, F(\alpha_{is}, \beta_{is} | Age, Sex, \Sigma) \tag{7}$$

$$e_{\sigma_j,\mathrm{FC}_j} = -\frac{\mathrm{FC}_j}{\sigma_j} \cdot \int \alpha_i L_j(\beta) \cdot (1 - L_j(\beta)) \, F(\alpha_{is}, \beta_{is} | Age, Sex, \Sigma) \tag{8}$$

4.2 Control Function

The control function approach is designed to address two forms of endogeneity. These sources of endogeneity stem from markdowns due to market power within industries. That is, I am assuming that firms have some market power derived from an oligopsonic market when hiring workers. This varies across occupations and states; some firms can exert more market power than others, depending on their occupation and geographic location. The second source of endogenity comes from job characteristics. While the markdown is a function of market power, it is also affected by job characteristics. The idea is that firms offering more desirable occupations have a stronger bargaining position against workers, and vice versa.

All occupations in my sample are blue-collar. Many of these occupations have both opportunities for self-employment and traditional wage labor. For example, a plumber could work for themselves, contract with a firm, or work as a wage worker. Within a geographic unit, workers may engage in these different types of employment at different rates. Take, for example, the trucking industry; the market power any individual firm can exert in hiring a commercial driver is lower relative to an occupation with a more centralized market. However, this will vary across states, as some have large employers or distribution centers, which alter the competitiveness of the labor market within each state.

Workers choose their occupation based on the expected wage they will earn, which is either the median wage or the 25th percentile, depending on whether they have prior experience. The wage the worker uses to choose his occupation is not the wage they would earn, but the wage they expects to earn. In essence, an omitted variable bias biases the worker's expected wage response due to the market power in the occupation they are considering. This market power puts downward pressure on wages they observe, both as a function of the working conditions in that occupation and unobservable factors. This endogeneity is addressed using the control function approach, which provides exogenous variation in the wage rate.

Since firms in more concentrated markets are able to exert more market power than those in less concentrated markets. The instruments I use are designed to leverage wage variation across markets and occupations that correlate with productivity while removing the variation due to local labor market features. The levels of markdowns will vary across states and occupations. However, the underlying productivity will be correlated.

Firms know their occupation's popularity among workers and leverage that knowledge as downward pressure on wages. Firms observe all features of their occupation, both those that are observable to a researcher through O*Net and those characteristics that are not observable. Since firms have information about unobserved job characteristics, they use this information to lower wages relative to less popular occupations. Explicitly, I model observed market wages as $w_{js} = \mathrm{mp}_j + \gamma_{js}$, the average marginal productivity as a constant and a markdown γ_{js} based on the market power and unobserved occupation characteristics.

Since I estimate the model using maximum likelihood, I use a control function approach with two sets of instruments to control for endogeneity. The first instrument is a Hausman instrument. I use the mean annual wage rate within each occupation of each bordering state as the instrument. This instrument exploits the correlation in marginal productivities across states, but removes variation due to local market factors. The idea is that the marginal productivity of workers is correlated across states, and the state-specific markdowns on unobservables are uncorrelated across states. Consider the case of unobservable training costs. Since this is a state-specific unobservable, the training cost is unobserved to the researcher for all non-truck-driving occupations. However, firms observe these costs and account for them when determining the wage rate. If the relationship between training programs and markdowns is uncorrelated between states, then the Hausman instrument should recover a consistent estimate of the coefficient on total compensation. Since truck drivers, plumbers, and other licensed trades in the sample are licensed at the state level, it is likely that the markets for training programs within states are uncorrelated with those in outside states. Formally, I am assuming all sources of market power between states s, s' satisfy:

$$E(\gamma_{js}\gamma_{js'}|x_j) = 0$$

The second set of instruments are the observed occupation characteristics. These characteristics are assumed to be stable in the sample period; that is, firms take the characteristics as given and are constant across all states. These characteristics are relevant to the firm's decision in determining the wage rate and the worker's decision in choosing that occupation. Controlling for these characteristics is important because all firms observe these characteristics. This causes a correlation in the markdowns across states since every firm uses the same information. I use both the Hausman instrument and observed characteristics as the instruments for the endogenous component of wages. Explicitly, I model the source of endogeneity as:

$$u_{ijks} = w_{js}\alpha_{is} + ||x_j, x_k||\beta_{is} + \xi_m + \varepsilon^1_{ijks} + \varepsilon^2_{ijks} \tag{9}$$

Where ε^1_{ijks} is the additively separable component of the error term correlated with wages. Let z_{js} be the Hausman instrument and x_j be the observed product characteristics. By using $w_{js} = \mathrm{mp}_j + \lambda_1 z_{js} + \lambda_2 x_j + \gamma^1_{js}$ to model wages, the new error term γ^1_{js} is uncorrelated with the endogenous component of wages. The utility model corrected for the endogenous component of wages then becomes:

$$u_{ijks} = w_{js}\alpha_{is} + ||x_j, x_k||\beta_{is} + \xi_m + \lambda\gamma^1_{js} + \varepsilon^2_{ijks} \tag{10}$$

Without the control function approach, the estimated wage elasticities would be biased. A downward bias would occur if occupations with lower productivity were more competitive, and an upward bias would occur if occupations with higher productivity were more competitive. Both the national and state-by-state models are estimated by maximum likelihood using the Hausman instrument and job characteristics in the control function.

4.3 Source of Variation between State and National Models

The National model exploits variation across states, workers, and occupations. The state model only uses variation across occupations within a state. This difference impacts how the control function works. In the case of the national regression, the variation in wages across states can introduce an additional correlation between market power and wages. At the state level, there is only the variation within the state, but across occupations, which can introduce correlation between market power and wages. This implies that states with less market power, characterized by more competitive labor markets, will have a lower correlation between market power and wages, as the observed wages in these more competitive states are closer to the true marginal productivity.

5 Estimation

The two models are estimated using simulation-assisted maximum-likelihood, in Python using the package Xlogit Arteaga et al (2022). The national regression is estimated using 38 of the 48 states in a cross-section. The second model is estimated state by state. I chose to estimate the two models separately due to the following tradeoff. I can use a GPU-assisted estimation using Google Colab, but that requires reducing the sample size to fit the data within the memory constraints Google enforces. GPU-assisted estimation is attractive because it speeds up the estimation process and allows more draws to compute the random coefficients, leading to more accuracy in computing the random coefficients. However, with this approach, I lose the efficiency gained from joint estimation. While, in principle, I could estimate the model jointly without GPU assistance, the downside of this approach is my software is not capable of handling a large number of observations and choices. So, I am forced to either estimate the model as a cross-section or a state-by-state repeated cross-section.

The first model is estimated as a cross-section entirely because of data constraints. The IPEDs data has considerable inconsistencies year over year in its reporting of CDL training programs. Because of this, I can't get a consistent sample within states across time. So, I use 2019 the most recent year, and estimate the model in cross-section.

Using the state-by-state approach, I can leverage my nine years of data and allow each state to differ in its coefficients. I do not have the same limitations with data consistency using OEWS and the CPS only.

6 Results

6.1 National Response

The regression results from Equation (10) at the national level are in table 3. This model is estimated for 38 out of the 48 states in the sample since I didn't observe the training cost data for ten states in 2019. The data consists of a single cross-section of 38 states in 2019. There are 8,347 workers in the model, and they choose among 42 occupations based on their initial occupation and observed wage rates. I report the conditional logit estimates and estimates without the control function, as well as the mixed logit with a control function.

The results indicate that workers are more likely to choose higher-paying occupations, all else being equal. Consistent with the endogeneity between wages and local market power, the coefficient on wages shrinks with the inclusion of the control function. Furthermore, since all estimates are positive, we find that workers prefer higher-paying occupations, but the naive regressions will overstate this effect.

There is substantial heterogeneity in preferences for distant occupations. On average, workers tend to prefer occupations that are more similar to their previous ones. This preference increases as workers age, and women are also substantially less willing to move to distant occupations. However, not all workers are averse to occupational differences. For example, consider a male with the average age in the sample (46.5), the mean response is a disutility of about 2.1 for a unit increase in distance between their current and past occupation. However, about 10% of the population of men age 46.5 will get positive utility for occupational differences. This increases as the age of the worker decreases; for male 21-year-olds, about 15% receive positive utility for occupational differences.

The ξ_m are the major occupation unobservables. These are the utilities of choosing an occupation within the major occupation code. The utilities of each major code are normalized to the

Table 3: Estimation Results

	Conditional Logit (1)	Mixed Logit (2)	Mixed Logit with CF (3)
Total Compensation	0.0227^{***}	0.0126^{***}	0.0100^{***}
	(0.0000)	(0.0015)	(0.0000)
Distance	-1.5996^{***}	-1.5549^{***}	-1.5404^{***}
	(0.0003)	(0.0903)	(0.0019)
Distance × Female		-0.3724^{***}	-0.3798^{***}
		(0.0522)	(0.0011)
Distance × Age		-0.0135^{***}	-0.0128^{***}
		(0.0018)	(0.0000)
Control Function			-0.0004^{***}
			(0.0000)
ξ_2 (Tradesman)	-0.8711^{***}	-1.0421^{***}	-1.0925^{***}
	(0.0012)	(0.0675)	(0.0015)
ξ_3 (Manual Labor)	-0.1906^{***}	-0.1326^{*}	-0.1449^{***}
	(0.0014)	(0.0760)	(0.0017)
ξ_4 (Service)	0.4890^{***}	0.2456^{***}	0.1877^{***}
	(0.0010)	(0.0603)	(0.0013)
ξ_5 (Production)	0.4463^{***}	0.3931^{***}	0.3627^{***}
	(0.0014)	(0.0783)	(0.0017)
ξ_6 (Transportation)	0.1244^{***}	0.1705^{***}	0.1609^{***}
	(0.0010)	(0.0583)	(0.0013)
ξ_7 (Sales)	1.0168^{***}	1.0371^{***}	1.1010^{***}
	(0.0009)	(0.0553)	(0.0012)
SD Total Compensation		9.80×10^{-6}	2.92×10^{-5}
		(0.0030)	(0.0001)
SD Distance		1.6308^{***}	1.6129^{***}
		(0.0358)	(0.0008)
Observations	8,347	8,347	8,347
Log-Likelihood	$-40,234$	$-37,128$	$-37,096$
First-Stage F-stat			42,150

Notes: Standard errors in parentheses. *** $p < .01$, ** $p < 0.05$, * $p < 0.1$. Sample: 38 states, 2019 cross-section. Total compensation is the wage net the training cost for truck driving. The control function uses Hausman instruments (neighboring state wages) and occupation characteristics. ξ coefficients are relative to agricultural workers.

utility of choosing agricultural labor. As such, it can reveal people's average preference for these occupations.[12]

This set of estimates is used to calculate the reduction of training cost counterfactuals. The other four counterfactuals are calculated using the state-by-state estimates. The mean wage elasticities are reported for truck drivers and the nine other occupations with the largest cross-wage elasticities. These estimates come from the state estimates.[13]. It is a selection of occupations that experience the largest decrease in market share when the wages of truck drivers rise. However, that does not mean these occupations are the largest source of new truck drivers. The initial share of workers could be small. The size of the move into trucking will be a combination of the elasticity and the size of the occupation from which the workers are moving. The elasticities reported here indicate the sensitivity of workers in these occupations to changes in truck driving wages.

Table 4: Estimated Average Elasticities

		1	2	3	4	5	6	7	8	9	10
Administrator/Manager	1	0.0275	-0.0004	-0.0038	-0.0012	-0.0001	-0.0002	-0.0001	-0.0001	-0.0002	-0.0001
Advanced Construction Workers	2	-0.0005	0.0140	-0.0006	-0.0006	-0.0002	-0.0005	-0.0003	-0.0003	-0.0006	-0.0004
Advanced Sales Worker	3	-0.0054	-0.0007	0.0274	-0.0016	-0.0002	-0.0002	-0.0003	-0.0002	-0.0004	-0.0002
Advanced Transportation Worker	4	-0.0014	-0.0005	-0.0013	0.0170	-0.0001	-0.0002	-0.0002	-0.0001	-0.0003	-0.0002
Bus and Taxi Drivers	5	-0.0002	-0.0003	-0.0004	-0.0002	0.0036	-0.0001	-0.0002	-0.0001	-0.0003	-0.0004
Construction Worker	6	-0.0003	-0.0008	-0.0003	-0.0004	-0.0001	0.0117	-0.0003	-0.0004	-0.0006	-0.0003
Courier	7	-0.0003	-0.0005	-0.0004	-0.0003	-0.0002	-0.0003	0.0061	-0.0002	-0.0004	-0.0014
Forestry Worker	8	-0.0002	-0.0005	-0.0003	-0.0002	-0.0001	-0.0005	-0.0002	0.0050	-0.0003	-0.0004
Rail Workers	9	-0.0003	-0.0006	-0.0004	-0.0003	-0.0002	-0.0004	-0.0003	-0.0002	0.0101	-0.0004
Truck Drivers	10	-0.0003	-0.0007	-0.0004	-0.0003	-0.0004	-0.0003	-0.0016	-0.0004	-0.0006	0.0062

Notes: The ten selected occupations with the highest cross-price elasticities to truck driving. Computed using the formula from equations (5) and (6).

In Table 4, I report the estimated average elasticities. Of the ten occupations, bus and taxi drivers have the smallest, and management has the highest own wage elasticity; truck drivers' own elasticity is about .0062. Somewhat unsurprisingly, couriers have the highest cross-wage elasticity with respect to a wage increase for truck drivers. Couriers deliver goods but do not drive vehicles large enough to qualify as truck drivers. The other occupations are a mix of construction, sales, and transportation-related fields.

The counterfactuals are computed using the 2019 data since this is the final year of data and is most germane to the current market. Each counterfactual modifies some component of 2019, for example, increases wages, and simulates the employment outcome as an increase in the number of

[12]The six major codes are trades, manual labor, service, production, and transportation workers listed in the table.
[13]The distribution of wage estimates is reported in the appendix in **??**

workers and a percent change in the number of workers. It reflects how many more truck drivers we would have had in 2019 if wages were higher, etc. Furthermore, it is not a simulation of the demand side. What employers are willing to pay is not captured by this model. For example, expanding the labor market of truck drivers by allowing 18-to 20-year-olds to enter the market is a classic supply shifter. All else equal, this will lower wages unless demand is perfectly elastic. However, I cannot compensate for the supply shift since I do not measure the employers' demand elasticities. Therefore, I assume the demand side elasticities are perfectly elastic; consequently, wages are constant. Thus, the counterfactual allowing 18-20-year-olds to enter the truck driving market should be an upper bound on the employment increase of that policy.

I simulated three of the five counterfactuals in two ways because the pecuniary counterfactuals can be reported as a percent or a nominal increase. Table 5 reports the results of these counterfactuals aggregated nationally. For the three pecuniary counterfactuals, I compute two cases, a 1%, and a $1,000 change. The first counterfactual is a wage bonus to both incumbent and potential truck drivers. This is the largest employment increase of all pecuniary benefits at about 6,600 or 21,300 new drivers, depending on the type of increase. From a firm's perspective, an increase in wages by $1,000 across the board is expensive. With approximately 3.5 million truck drivers in the United States, the increase in the wage bill resulting from such a policy would be substantial. Another potential tool is to increase the wages of a new entrant. The sign bonus row is an increase in wages for a new entrant, but not incumbents. This policy is attractive for firms because it does not increase the wage bill for all workers, just new entrants. Simultaneously, it has an effect size about 30% smaller than the previous counterfactual at a much lower cost. This tradeoff may be attractive to firms since it gets most of the benefit with a much lower wage bill. The final pecuniary counterfactual is a decrease in training cost as either a 1% or $1,000 transfer. This policy has a relatively small impact compared to the wage and sign bonus. Since training costs are substantially smaller than wages, a 1% decrease in training cost works out to be a $24 scholarship at the mean cost, so it is somewhat unsurprising the effect size is so small. However, for a $1,000 increase, the impact is still about half of the impact of a $1,000 signing bonus. The two remaining counterfactuals are non-pecuniary and adjust the age profile of the 2019 labor market. The first policy is to allow 18-20 year olds to enter the truck driving market. The policy increases to about 50,000 drivers nationally. Two things are driving this result. First, younger people are likelier to

Table 5: Aggregate Responses

	1% Increase		$1000 Increase	
	Employment	% Change	Employment	% Change
Wage Response	6,614	0.17%	21,292	0.54%
Sign Bonus	4,479	0.11%	15,469	0.39%
CDL School Cost	480	0.01%	7,151	0.20%
18-20 Year Olds	46,481	2.44%	46,481	2.44%
Aged 5 Years	18,773	0.48%	18,773	0.47%

Notes: These are the aggregate responses for each of the counterfactuals. All results were computed using the state-by-state estimates except for the CDL School cost reduction.

switch to different occupations than older groups. Therefore, all younger workers are more likely to switch occupations. And since, by legislation, all 18-20-year-olds are not engaged in trucking as an occupation, they are not more likely to switch out of truck driving because they never occupied the profession to begin with. This results in a large share of them leaving their incumbent occupation, making them a good source of new drivers. To illustrate this, suppose that older workers were by law unable to become truck drivers. Since these workers are extremely averse to new occupations relative to other groups; it would be much harder to coax them from their incumbent occupation. Secondly, there are a large number of 18-20-year-olds in the blue-collar labor market, with about 3.99 million predicted using the CPS 2019. Since that pool of workers is so large, recruiting a modest share, about 1.6% on average, results in considerable increases in truck drivers. Almost all of the increase comes from the expanding labor pool since the increase in market share of truck driving only increases the market share of truck drivers by an extremely small share. The final counterfactual simulates what happens when the age distribution increases by five years and workers are required to retire at 70. Aging the workforce decreases the moves between occupations since occupational differences cause more disutility as workers age. Aging the workforce results in about 13,400 more truck drivers. This is largely a consequence of higher retention in the truck driving profession; people who were more likely to move occupations when they were younger are less likely as they age.

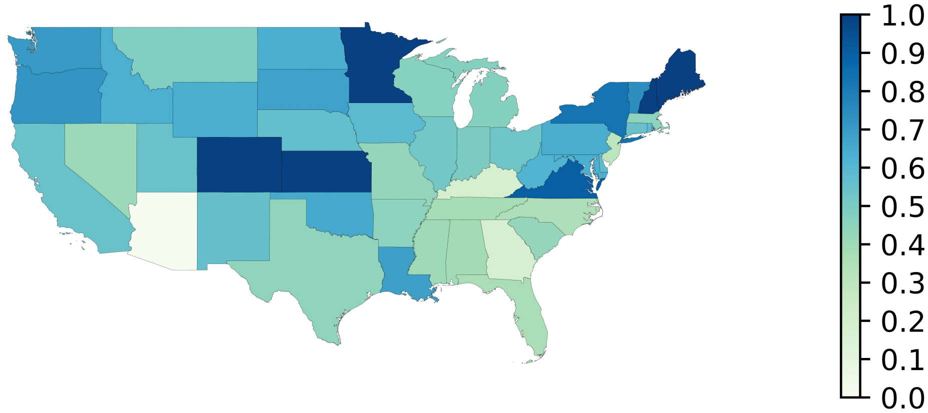
6.2 State Responses

The next set of results focuses on four counterfactuals using the state-specific responses. The progression of results follows the same structure as the national response table, that is, wage increase, sign bonus, 18-20-year-olds, and shifting the age distribution. These results encapsulate which states are the biggest contributors to the increase in truck driving employment. All pecuniary counterfactuals are reported for a $1,000 change.

Both a thousand-dollar increase in truck driver wages and a sign bonus result in the following geographic pattern. Figure 3 reports the geographic distribution of a wage increase. The figure's legend records the percent increase in truck drivers by state for a thousand-dollar wage increase. The geographic pattern is equivalent to the signing bonus counterfactual. The only difference between the two counterfactuals is the size of the impact. The darker blue colors are associated with a larger response; the lighter the color, the smaller the employment increase. Minnesota, Maine, Kansas, Colorado, and New Hampshire have the largest percentage increase in drivers within their state. Arizona is unique, with a response very close to 0. The Deep South as a group, except Louisiana, has reasonably small responses to wage increases. The Upper Midwest and Great Plains regions have higher responses than the southern regions. The geographic pattern is generated by the differences in the wage coefficient across states. States differ in the average impact of wage changes and the standard deviation of workers' wage responses. States with the largest average wage response have the largest elasticities in Figure 3.

Allowing 18-20 year olds to become truck drivers has an interesting geographic pattern. Figure 4 Shows the share of 18-20-year-olds that choose truck driving as an occupation. This counterfactual demonstrates the utility of a mixed logit approach since the state responses depend on the distribution of prior occupations, younger workers' preferences for different occupations, and the gender and age distributions within people's previous occupations. States where the 18-20-year-old demographic are more likely to be incumbents in occupations that are most similar to truck driving will simultaneously be the states that get the biggest gains from allowing 18-20-year-olds to enter the market. Essentially, this figure demonstrates the willingness of 18-20-year-olds to choose truck driving as a profession by state. The states where the largest share of 18-20-year-olds choose truck driving are Missouri, Arkansas, Georgia, North Dakota, Kentucky, Idaho, and Wyoming. Unlike

Figure 3: $1,000 Increase in Annual Earnings



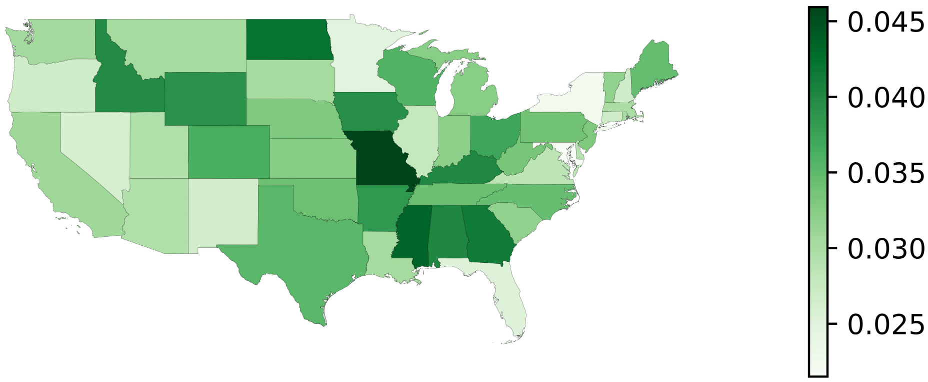
Notes: Percent change in the number of truck drivers for a $1,000 annual wage increase. Calculated from recomputing the market share of truck drivers after increasing the wage by $1,000.

the wage counterfactuals, the Southern states get large increases in truck drivers.

However, states also differ in the share or number of 18-20-year-olds in their workforce; states with the largest share of younger workers will get the largest increases in truck driving by letting them enter the market, even if a relatively small number of 18-20-year-olds would choose truck driving. Figure 4 ignored the size of 18-20-year-olds in the market by state; for example, a state like Texas has a modest share of 18-20-year-olds predicted to choose truck driving but has one of the largest increases in the number of new truck drivers as demonstrated in Figure 5. Texas and California dominate the geographic pattern because these states have a large number of 18-20-year-olds. Ohio, Florida, and Pennsylvania also see moderately large increases. The low-population states in the Great Plains region see small increases relative to the share of 18-20-year-olds willing to choose truck driving in those states.

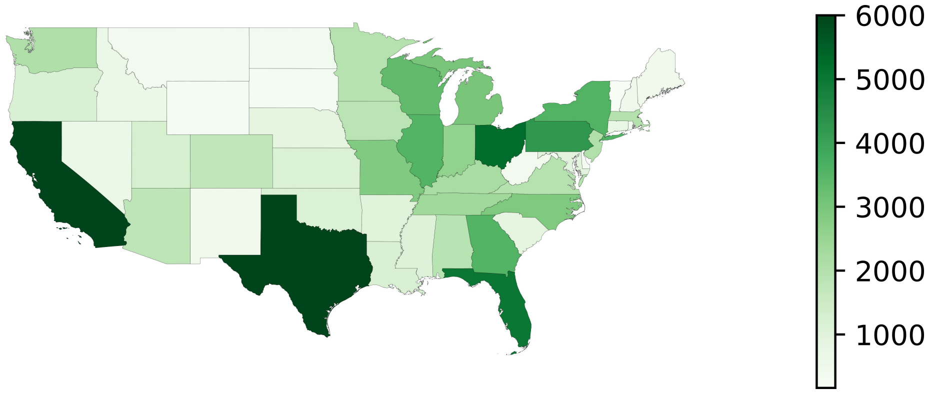
The final counterfactual is calculated by increasing the age distribution by five years and retiring workers 70 years and older. Figure 6 reports the percent change in truck driving employment by state. This counterfactual is interesting because some states experience a net decline in truck drivers, although the plurality of states will see increases. Nevada, New York, Maryland, South Carolina, and Massachusetts see declining numbers of truck drivers as the age distribution rises. This distribution of prior occupations in each state is important for determining whether a state sees an increase or decrease in the share of truck drivers. Individuals are less likely to switch to more

Figure 4: Share of 18-20-Year-olds that choose Truck Driving



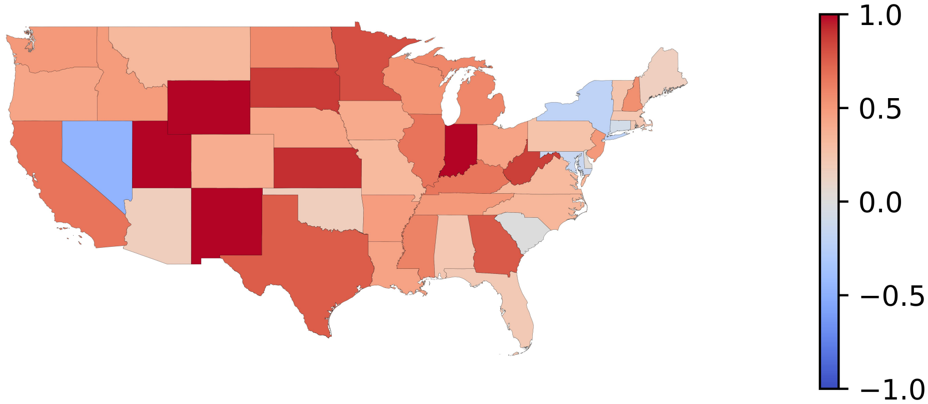
Notes: Share of 18-20-year-old workers that choose to enter the market for truck drivers. Calculated from computing the market share of 18-20-year-olds that choose truck driving as an occupation by state.

Figure 5: Increase in truck drivers allowing 18-20-year-olds enter the market



Notes: Number of 18-20-year-old workers that choose to enter the market for truck drivers. Calculated from computing the market share of 18-20-year-olds that choose truck driving as an occupation by state and multiplying that by the total number of 18-20-year-olds in that state. The largest value is truncated at 6,000 since Texas contributes a large share relative to other states' effect sizes.

Figure 6: State heterogeneity in the share of Truck Drivers for a five-year increase in the age distribution



Notes: This figure shows the percentage point increase in the number of drivers after aging the workforce by five years. I retire all workers that exceed the age of 70, but I keep the total labor force size the same. The total employment in the blue-collar market is the same before and after the counterfactual change.

distant occupations as they age. As the workforce ages, if a greater share of the occupations in these states are more distant from truck driving, the likelihood of people switching to truck driving falls. Put another way, the choice probabilities of occupations most similar to the incumbent occupation rise as the age distribution rises. Then, if there are a sufficiently large number of people in those occupations that are most similar to truck driving, the share of drivers will increase.

The declines also come from retirement. Since I force workers over the age of 70 to retire, all the drivers that exceed 70 will be dropped from the market. If the gains from retaining more workers are less than retirements or the distribution of prior occupations is sufficiently different from truck driving, we will observe a net decline in movers from other occupations. The red states are all states that see increases in the number of truck drivers as they age. New Mexico, Utah, Wyoming, and Indiana are the states that see the biggest increases in the share of drivers.

The final set of results tabulates the top ten states as measured by the employment increases for each of the counterfactuals. These employment increases do not lend themselves well to choropleths since a few states dominate the employment increases due to either their elasticity or large population. Table 6 contains the four state counterfactuals, with the two pecuniary counterfactuals computed for a $1,000 increase.

California, Texas, and New York are all very large states in terms of employment; typically,

Table 6: State Decomposition: Top 10 States by $1,000 Increase

State	Wage	State	Sign Bonus	State	18-20	State	Aged 5 Years
California	2,492	California	1,768	Texas	8,962	California	2,955
Texas	1,647	New York	1,365	North Carolina	3,814	Texas	2,860
New York	1,567	Texas	1,191	Florida	3,018	Georgia	1,333
Colorado	1,048	Colorado	754	Massachusetts	2,480	Indiana	1,175
Pennsylvania	1,000	Pennsylvania	677	Illinois	2,365	Illinois	971
Ohio	947	Ohio	655	Pennsylvania	2,138	Ohio	790
Minnesota	874	Minnesota	619	Missouri	2,069	Michigan	585
Illinois	797	Virginia	578	California	1,825	New Jersy	557
Virginia	779	Illinois	572	Ohio	1,735	Wisconsin	524
Florida	738	Florida	555	Michigan	1,560	Minnesota	499

Notes: This table tabulates the biggest contributors to each counterfactual by state. California, Texas, and New York tend to dominate due to their large populations

they contribute many new truck drivers in each counterfactual scenario. In the case of the wage increase and sign bonus counterfactual, the top 10 states contribute over 50% of the national rise in truck drivers. Allowing 18-20-year-olds to enter the truck driving market leads to the largest increase in truck drivers. Again, Texas and California are big contributors. However, some smaller states also contribute to the increase in drivers. North Carolina, Florida, Massachusetts, Illinois, and Pennsylvania as a group contributed about 30% of the increase in truck drivers. Finally, the increases from shifting the age distribution follow approximately the same pattern with the loss of New York as a big contributor. California and Texas are the top two states. New Jersey and Indiana are also large contributors, which follow from Figure 5.

7 Consequence for other Occupations

The total number of workers stays constant in each counterfactual except for allowing 18-20-year-olds into the market, where the interpretation differs slightly. Therefore, for 4 of the five counterfactuals, the gains for one occupation mean another occupation is losing some percentage of its workforce. Consequently, any policy to increase the number of drivers will negatively impact other occupations, a key result relevant for policymakers. For example, bus drivers are one of the most sensitive occupations to changes in truck driving. By increasing the number of truck drivers, we will necessarily decrease the number of bus drivers. Understanding the tradeoffs between having more drivers and the consequences for other occupations is important for deciding whether a policy should be pursued. I report the top 10 occupations by the number of jobs lost for each counterfactual except the 18-20-year-old counterfactual. Table 7 reports the results for the wage and sign bonus.

Table 7: Occupational Impact for Increasing Wages in Truck Driving

Wage Increase ($1,000)	Δ Employment	Sign Bonus ($1,000)	Δ Employment
Courier	-4,593	Courier	-2,172
Bus and Taxi Drivers	-1,171	Rail Workers	-807
Advanced Construction Workers	-1,079	Advanced Construction Workers	-765
Forestry Worker	-1,062	Bus and Taxi Drivers	-745
Rail Workers	-1,035	Housekeeping/Janitorial Worker	-701
Housekeeping/Janitorial Worker	-890	Advanced Sales Worker	-673
Construction Worker	-844	Construction Worker	-648
Advanced Sales Worker	-796	Forestry Worker	-574
Material Movers	-563	Material Movers	-442
Advanced Transportation Worker	-529	Extraction Worker	-433

Notes: Table reports the impacts of increasing wages in truck driving by $1,000. Reports the most workers lost by occupation using the state-by-state estimates.

Table 7 is broken into two parts: the wage increase on the left and the sign bonus on the right. The numbers are reported as the net decline in the number of workers nationally in each occupation as the wage of truck drivers rises. As reported in Table 4, Couriers have the highest cross-wage elasticity relative to other occupations in the sample. As such, they see the biggest decline in their workforce as the number of truck drivers increases. Most occupations are unsurprising; the linkages from truck drivers to the Bus and Taxi Drivers and Couriers, Construction, Rail, Material Movers, and transportation workers are evident. These occupations are related to truck drivers, either because truck drivers enable these workers to complete their tasks or they already have a specialized license similar to that of the CDL required for truck drivers. Forestry and Housekeeping/Janitorial workers are, at first observation, surprising members of this group of occupations. However, there is a large number of janitorial workers in the sample, so even if a small share of them leave to become truck drivers, the impact is fairly large. Forestry workers may see a decline in workers because of their primarily rural status. Truck driving is a fairly good occupation in many rural areas because the job builds in commuting, and while working, a driver is on the road to deliver or pick up a load, not traveling between home and work. Therefore, it may be an easier transition for forestry workers to become truck drivers, although this claim needs further investigation. The results for a sign bonus are qualitatively similar to that of the wage increase with one major difference. The advanced transportation workers fall from the tenth spot and are replaced by extraction workers.

The extraction workers are coal, oil, and natural gas extraction workers. They work in a variety of contexts, extracting fuel for the US.

The second counterfactual I consider allows the workforce to age by five years. Because it doesn't increase the number of workers, not all occupations benefit. Any change in the workforce constitutes a reshuffling of workers across occupations. This counterfactual shows what happens if there is no growth in the blue-collar labor market, and the workforce continues to age. Table 8 reports the results for an aging workforce.

Table 8: Aging Counterfactual: Workforce Average Age Increases by 5 Years

Top Occupations (H-L)	Δ Employment	Bottom Occupations (H-L)	Δ Employment
Office Clerk	79,467	Rail Workers	-62,746
Sales Worker	77,676	Advanced Sales Worker	-45,052
Production/Operation Workers	75,829	Advanced Transportation Worker	-39,030
Food Preparation Worker	55,190	Gaming Service Worker	-33,613
Construction Worker	49,198	Printing Workers	-32,481
Housekeeping/Janitorial Worker	47,188	Textile/Shoe Workers	-30,939
Emergency Service/Criminal Justice	38,418	Extraction Worker	-30,274
Transportation Clerk	36,135	Laundry Workers	-29,158
Service Worker	34,952	Energy Installer	-26,120
Material Movers	33,225	Funeral Workers	-24,627

Notes: Table reports the impacts of increasing ages of the workforce by 5 years. Reports the most workers lost and gained by occupation using the state-by-state estimates.

As the workforce ages, there are two components of the mode to consider. First, aging makes workers less willing to move to different occupations. The disutility for moving from one occupation to another increases as workers age, which implies that as the workforce ages, we will observe less moving between occupations. In effect, the labor market is crystallizing. Secondly, occupations differ in their retention and entrance rates. There are occupations where a large share of the workforce comes from new entrants; these are occupations that will experience the biggest declines as the labor market ages. The occupations with the highest retention rates are going to see the biggest increases in the number of workers because workers who would have left had they been younger are no longer willing to leave the occupation. In essence, high retention occupations see increases as the labor market ages, and high entrance rate occupations see the biggest declines. Table 8 records the top and bottom ten occupations by the number of workers the occupation gains or loses. Notably, 61% of occupations see a decline in the number of workers as the labor market ages. The other notable fact is that truck driving is not in the top or bottom ten occupations.

8 Choices of Young Workers

The counterfactual allowing 18-20-year-olds to choose truck driving resulted in an additional 46,481 workers becoming truck drivers. This counterfactual is computed by comparing the world if 18-20-year-olds are eligible for trucking to if they are ineligible. Once truck driving is an option for young workers, I can compute the occupations that get the largest change from allowing truck driving into the choice set. Table 9 records the change in 18-20-year-olds by occupation by including truck driving in the choice set.

Table 9: Allowing 18-20-Year-Olds to choose Truck Driving

Top Occupations (H-L)	Δ Choice	Initial Choice	Occupations (H-L)	Δ Choice	Initial Choice
Food Service Worker	1,103	251,988	Courier	-9,576	33,269
Bus and Taxi Drivers	699	22,281	Sales Worker	-5,728	619,556
Production/Operation Workers	501	89,218	Construction Worker	-3,119	63,280
Event Manager	383	21,840	Rail Workers	-2,954	42,204
Gaming Service Worker	56	44,028	Advanced Construction Workers	-2,148	34,831
Repair Worker	-86	19,607	Housekeeping/Janitorial Worker	-2,123	94,586
Advanced Electronics	-92	8,718	Forestry Worker	-1,555	18,803
Laundry Workers	-133	25,034	Energy Installer	-1,470	19,778
Media Technician	-138	25,533	Office Clerk	-1,452	93,967
Personal Appearance Workers	-188	72,806	Extraction Worker	-1,360	25,663

Notes: Table reports the impacts of allowing 18-20-year-olds to choose truck driving as an occupation. The table reports the most workers lost and gained by occupation using the state-by-state estimates.

While this is notionally similar to the other counterfactuals, the interpretation is slightly different. The numbers in the table represent a recomposition of the choices young workers make, whereas the tables in the prior section are a recomposition of the entire labor market. First, the occupations that young workers choose are different from the occupations that more mature workers make. The bulk of young workers choose occupations like sales, food preparation or service, and material moving. Interestingly, allowing young workers to choose truck driving has a diverse impact on these popular occupations. For example, more workers choose food service, but fewer choose sales. This likely has a lot to do with the demographics of young sales workers relative to food service workers and the job similarities. The majority of occupations experience a loss in workers. All but five occupations have a decline in young workers. Unsurprisingly, couriers would see a reduction of 9,576 young workers when truck driving is an option. In each counterfactual, couriers are the most willing group to switch to truck driving.

However, these results are independent of the scale of the occupation. The second column records the number of 18-20-years that choose each occupation when truck driving is not in the choice set. In terms of decline, it is clear that a large number of young couriers would switch to truck driving, given the opportunity, and that would cause a relatively large decline in the number of young workers in that occupation. This constitutes a 29% reduction in young couriers. In contrast, the 5,700-worker decline in sales is a much smaller share of the total workforce (about 1%) and likely would have a small impact on the labor market in that occupation. As for the rest of the occupations, the results are much less extreme. Most occupations lose between $\pm 0 - 8\%$ of their workforce. And only 6 of the occupations lose 5% or more. For most occupations, a few workers switch to truck driving, but this would likely have minimal impact on the labor market for these workers.

9 Conclusion

I used four data sets to estimate a mixed logit model of occupational choice to compute five counterfactuals. Given the large concern of a truck driving shortage, my five counterfactual experiments address some of these concerns. The wage and sign bonus counterfactual demonstrates that workers are responsive to wage increases. A $1,000 increase in all wages nets about 21,100 more drivers nationally, and a $1,000 sign bonus nets about 15,500. While these increases are small in the context of the approximately 3.5 million drivers in the United States, it is clear that workers respond to higher wages. An inelastic labor supply does not imply a shortage of workers. Firms may wish that smaller wage increases would coax enough workers into their occupations to cover the perceived shortfall. However, the small responses do not imply a shortage. A sign bonus is an attractive tool for firms to use. A larger bonus will induce more workers to switch occupations at a reasonable cost; however, retaining them becomes a challenge. Presumably, such a policy would lead to a higher wage bill as firms pay more to retain their workers.

The aging of truck drivers has often been cited as a potential cause of the shortage. In my counterfactual, I age the distribution of workers by five years and retire all workers older than 65. Surprisingly, this counterfactual showed increases in the number of truck drivers in most states. Only Nevada, New York, Maryland, and Massachusetts experience declines in the number of drivers as their workers age. Of course, if all drivers get older and no young drivers enter the market, the number of truck drivers will fall.

In the final counterfactual, I allow 18-20-year-olds to enter work as truck drivers. A policy of allowing 18-20-year-olds to work as truck drivers results in the largest increase in drivers. Currently, DOT regulation forbids 18-20-year-olds to drive commercial trucks across state lines. The Bipartisan Infrastructure Bill financed a trial program to authorize young workers to apprentice and work as truck drivers. My counterfactuals predict a nationwide increase of about 97,000 new drivers if 18-20-year-olds are allowed to enter the market. This result ignores the concerns about safety associated with allowing younger drivers to work as professional drivers. More research into the unintended consequences of allowing young drivers to enter the market is needed.

Finally, I compare which occupations see the biggest losses in the pecuniary counterfactuals. I find that those occupations that are most similar in proximity or skills to truck driving see the biggest declines in their workforce. This result is consistent in the counterfactual testing of the aging labor market. Finally, decomposing the impact of allowing 18-20-year-olds to enter the market has a wide variety of impacts across occupations. Like the other counterfactuals, couriers lose the most workers to truck driving, likely due to the occupational similarities.

On balance, the concern about a shortage is understandable but is likely unwarranted. This paper shows that the estimated wage elasticities are low, and inducing new truck drivers to enter the occupation is expensive. In this regard, it is understandable how firms would feel as if there is a shortage of workers. However, a low elasticity does not imply a market failure; it is more likely that firms are unwilling to pay a sufficiently high wage to induce the number of workers to drive that the firm perceives is necessary. In the case of truck driving, in particular, a regulatory burden exists but is significantly lower compared to other occupations, such as nursing. And it is not obvious what structural barrier would keep the occupation in perpetual shortage. The one most obvious structural barrier, alleviated by allowing 18-20-year-old workers to enter the market, provides a new pool of workers that can be hired to meet demand for transportation services. This is the largest increase in new drivers, but as I mentioned earlier, there are clear safety concerns that need to be addressed. The cost and benefits of such a policy are not straightforward. Finally, this paper suggests that recruiting workers from a handful of occupations similar to trucking would result in more workers. Those workers who are incumbent in occupations that are most similar to truck driving switch at the highest rates to truck driving,

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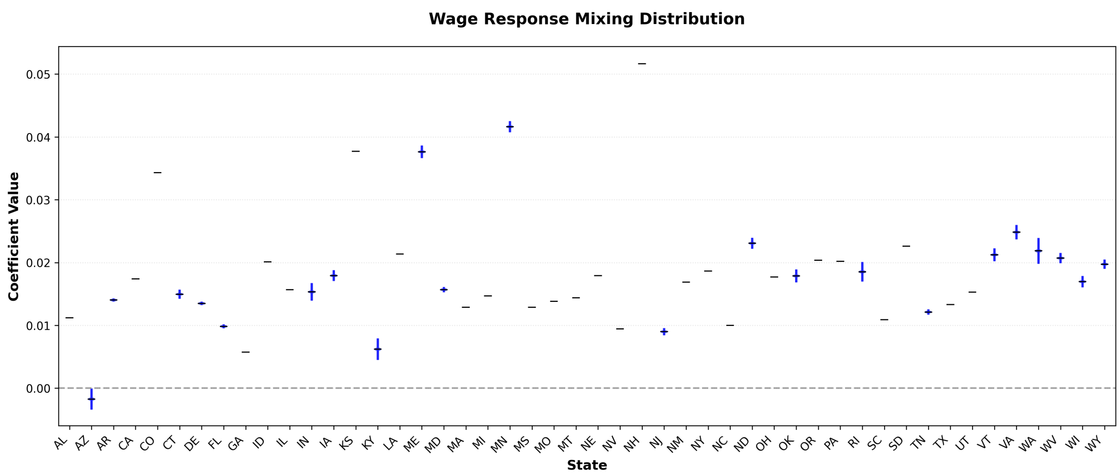
A State Mixing Distributions

Since each regression was run separately for each state, I have 48 separate regressions, each with several variables. The full regression suite is available on request. I report the estimated mixing distributions. These are related to the wage elasticities presented in 3, but provide a more precise summary, which compromises readability.

Figure Appendix-1: State-Specific Wage Coefficient Distributions



Notes: Mixing distributions from 48 state-specific regressions.

From Appendix-1, we can see that all states except Arizona have positively estimated wage elasticities. Most estimated standard deviations are low and are not statistically different from zero. A few states, like New Hampshire and Minnesota, have high estimated elasticities. This is evident from the wage response results decomposed by state

B First-Stage Regression Results

Table Appendix-1 reports the first stage estimates. These are regression estimates used to compute the control function used in both the state and national regressions.

Table Appendix-1: First Stage Regression Results

	(1) Wage
IV	0.940***
	(0.002)
Consequence of Error	220.722***
	(79.288)
Contact With Others	196.466**
	(87.016)
Degree of Automation	202.612***
	(77.019)
Exposed to Hazardous Conditions	346.957***
	(48.495)
In an Enclosed Vehicle or Equipment	365.355***
	(70.014)
Outdoors, Exposed to Weather	−261.923***
	(60.736)
Spend Time Using Hands	−508.635***
	(48.435)
Time Pressure	242.525***
	(92.716)
Constant	357.959
	(600.344)
Observations	59,190
R-squared	0.865
Adjusted R-squared	0.865
F-stat	42,150

Standard errors in parentheses

*** $p<0.01$, ** $p<0.05$, * $p<0.1$

C Missing States

The missing states in the national regression are: Colorado, Connecticut, Louisiana, Maine, Massachusetts, Nevada, New Jersey, Rhode Island, Vermont, and Wyoming.

The counterfactuals are computed assuming the national response elasticity estimated applies to all 48 contiguous states, so the estimated increase in drivers is comparable across counterfactuals.

D Data Construction and Variable Definitions

Table Appendix-2 presents a sample of the crosswalk between the O*Net, CPS, and my own codes. It includes only the occupations in the transportation industry.

Table Appendix-2: Trucking Occupation Code Crosswalk

Original Occupation	**Mapped Category**	**General Category**
Transportation, Storage, and Distribution Managers	Advanced Transportation Worker	Transportation Worker General
Logisticians	Advanced Transportation Worker	Transportation Worker General
Air Traffic Controllers and Airfield Operations Specialists	Advanced Transportation Worker	Transportation Worker General
Transportation Inspectors	Advanced Transportation Worker	Transportation Worker General
Bus and Ambulance Drivers and Attendants	Bus and Taxi Drivers	Transportation Worker General
Taxi Drivers and Chauffeurs	Bus and Taxi Drivers	Transportation Worker General
Couriers and Messengers	Courier	Transportation Worker General
Postal Service Mail Carriers	Courier	Transportation Worker General
Motor Vehicle Operators, All Other	Large Vehicle Operators	Transportation Worker General

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Table Appendix-2 – continued from previous page

Original Occupation	**Mapped Category**	**General Category**
Industrial Truck and Tractor Operators	Large Vehicle Operators	Transportation Worker General
Refuse and Recyclable Material Collectors	Large Vehicle Operators	Transportation Worker General
Material moving workers, nec	Large Vehicle Operators	Transportation Worker General
Postal Service Mail Sorters, Processors, and Processing Machine Operators	Material Movers	Transportation Worker General
Stock Clerks and Order Fillers	Material Movers	Transportation Worker General
Mail Clerks and Mail Machine Operators, Except Postal Service	Material Movers	Transportation Worker General
Supervisors of Transportation and Material Moving Workers	Material Movers	Transportation Worker General
Transportation workers, nec	Material Movers	Transportation Worker General
Conveyor operators and tenders, and hoist and winch operators	Material Movers	Transportation Worker General
Crane and Tower Operators	Material Movers	Transportation Worker General

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Table Appendix-2 – continued from previous page

Original Occupation	**Mapped Category**	**General Category**
Dredge, Excavating, and Loading Machine Operators	Material Movers	Transportation Worker General
Laborers and Freight, Stock, and Material Movers, Hand	Material Movers	Transportation Worker General
Machine Feeders and Offbearers	Material Movers	Transportation Worker General
Packers and Packagers, Hand	Material Movers	Transportation Worker General
Locomotive Engineers and Operators	Rail Workers	Transportation Worker General
Railroad Brake, Signal, and Switch Operators	Rail Workers	Transportation Worker General
Railroad Conductors and Yardmasters	Rail Workers	Transportation Worker General
Subway, Streetcar, and Other Rail Transportation Workers	Rail Workers	Transportation Worker General
Managers, nec (including Postmasters)	Transportation Clerk	Transportation Worker General
Cargo and Freight Agents	Transportation Clerk	Transportation Worker General
Dispatchers	Transportation Clerk	Transportation Worker General

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Table Appendix-2 – continued from previous page

Original Occupation	**Mapped Category**	**General Category**
Postal Service Clerks	Transportation Clerk	Transportation Worker General
Shipping, Receiving, and Traffic Clerks	Transportation Clerk	Transportation Worker General
Driver/Sales Workers and Truck Drivers	Truck Drivers	Transportation Worker General